

Per-task gate threshold fitting

Qwen2.5-Omni-7B on OmniPro Online — the full record:
approach, results, figures, findings, and what none of it shows

1 September 2026

One-paragraph summary. A streaming assistant must decide *when* to speak. It decides by thresholding p_{hit} , a probability read off the frozen model’s logits. This study fitted that threshold *per task* over two grid passes on a frozen 135-video subset, then ran the complete 2,700-sample benchmark. **The fit does not hold up, and the reason is measurable:** across all nine tasks the AUC of p_{hit} against event-adjacency is **0.394–0.526** — chance — so a threshold controls *how often* the system speaks and never *when*. Nine fitted thresholds do not beat one global threshold outside noise. The negative result is established four independent ways and is the contribution.

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1 What the system does and what is being fitted

A video plays. The model watches and listens as it arrives. At some moment it should *speak* — report a count, name a state change, alert on an event. The question is not *what* to say; it is **when**.

The deciding machinery is small. Once per tick the system asks the frozen model a yes/no question — “do we have enough information / has the event happened?” — and reads $P(\text{true})$ off the logits. Call it p_{hit} . If p_{hit} crosses a threshold, the system speaks.

Nine tasks share that mechanism, and the motivating guess was that they want *different* thresholds: a task that should alert once per video is not a task that should narrate continuously.

The benchmark. OmniPro Online: 9 tasks $\times$ 300 videos = **2,700** samples, 24.1 s–594.6 s, 65% requiring audio. **The metric.** `time_f1`: an emission counts if it falls within ± 3 s of a ground-truth trigger, matched greedily one-to-one. It needs no LLM judge, which is why every decision here is made on it.

The gate is three coupled knobs, not one: the threshold, the mode (`edge` = fire on a rising crossing, `level` = fire on every tick above), and the refractory (seconds of enforced silence after firing). **Only the threshold was refitted.** The other two are inherited unchanged from the vision-only predecessor. This is the study’s largest structural limitation and it recurs throughout.

2 Why the fit was necessary: the knob was disconnected

An audit of the shipped system found the threshold was not in the firing path at all. Under `gate_strategy="hysteresis"` the test that actually decided firing was `p_hit >= gate_high_thr`, a fixed global **0.5**; the per-task `hit_threshold` only decided whether an answer was *decoded*.

Across a completed 2,700-sample run — **581,403 ticks, 50,433 fires** — **not one fire occurred below $p_{hit} = 0.500$** . The four tasks configured at or below 0.5 were all silently running at 0.5. `task_gate_modes` and `task_refractory_s` were read **nowhere at all**.

The failure was **silent**: it produced a finished run and a number you could put in a table. It also means the “shipped” thresholds were never actually tested at their stated values — so this study’s baseline is itself a fiction that had to be re-measured.

2.1 A second defect, found while the fit was running

p_{hit} is high on the **very first tick** of every video: the model answers from the prompt before it has seen anything (≥ 0.05 for 98–100% of samples, median 0.34–0.64). Seeding the gate’s `prev_above` to **False** made that first tick a rising edge, so a long-refractory task spent its single allowed emission at $t \approx 1$ s against ground truth at 26–98 s.

The fix is definitional — an edge requires a predecessor, so `prev_above` starts **None**. Offline best time-F1 rose $0.0535 \rightarrow 0.0900$ (`snapshot_counting`) and $0.0636 \rightarrow 0.0751$ (`instant_event_alert`); short-refractory tasks moved ± 0.001 . **The size of the effect tracks the refractory**, which is the practical proof that the three knobs cannot honestly be fitted one at a time.

3 Approach

stage	what it does	cells	sample-evals	status
pass 1	wide grid, 10 thresholds $\times$ 9 tasks	94	1,410	complete
pass 2	refine 5 points around each pass-1 winner	45	675	complete
stage 3	the full benchmark at the fitted values	1	2,700	95%

The frozen subset. Passes 1–2 run on a fixed 5% all-audio subset — **15 videos per task, 135 total** — chosen once and never changed. Freezing it is what makes the two passes comparable to each other and to the bootstrap auditing them.

Protocol rules behind every number here:

- **Bootstrap resamples *videos***, never ticks or events. Ticks within a video are serially correlated through one shared KV cache; resampling them would manufacture confidence that is not there.
- **Paired, and it replays the selection rule.** Every threshold is scored on the *same* resampled video set, and the rule is re-run inside each draw rather than an argmax taken — so the audit measures the stability of *the procedure*, not of one number.

- **Re-selection is always divided by chance.** A 10-cell grid has a 10% chance rate; a 5-cell grid, 20%. Comparing raw rates across grid sizes is meaningless, and doing so would have made pass 2 look better than pass 1.
- **A 0.03 noise band.** The eval is not bit-reproducible; run-to-run variance on one fixed config was measured at F1 0.255 vs 0.051.
- **Content metrics are withheld, never estimated** when the judge is unreachable. A blank cell is a fact; a guessed cell is a lie that reaches a paper.

4 The fitted thresholds

task	shipped	fitted	gap vs rival	grid span	re-sel	chance
cumulative_counting	0.925	0.6167	−0.0017	0.048	48%	20%
dedup_counting	0.992	0.5500	+0.0243	0.055	70%	20%
event_narration	0.100	0.6500	+0.0037	0.023	74%	20%
explicit_target_grounding	0.500	0.7167	+0.0039	0.065	47%	20%
instant_event_alert	0.450	0.4500	+0.0645	0.065	63%	20%
realtime_state_monitor	0.800	0.4500	−0.0093	0.029	38%	20%
semantic_condition_alert	0.980	0.2167	+0.0091	0.066	42%	20%
sequential_step_instruction	0.010	0.1500	+0.0025	0.022	41%	20%
snapshot_counting	0.985	0.3833	+0.0667	0.133	56%	20%

The values move a long way from the shipped ones — but see §5 before treating any as a discovery. `instant_event_alert` did not move: it scored **0.000 at every threshold** in pass 1, so the rule correctly flagged it FLAT and kept the shipped value rather than picking a winner out of noise.

5 Findings

5.1 Finding 1 — p_{hit} does not rank event-adjacent ticks above quiet ones

This governs everything else. The ROC of per-tick p_{hit} against a ± 3 s positive label, now measured on **all nine tasks** with paired-over-video bootstrap CIs:

task	ticks	% positive	AUC [95% CI]
dedup_counting	4596	13.6	0.394 [0.35, 0.45] [†]
semantic_condition_alert	3323	10.9	0.431 [0.36, 0.53]
event_narration	4034	11.3	0.454 [0.41, 0.51]
sequential_step_instruction	3300	22.9	0.454 [0.40, 0.51]
realtime_state_monitor	2882	18.7	0.455 [0.39, 0.52]
cumulative_counting	2833	16.2	0.469 [0.43, 0.53]
explicit_target_grounding	2947	4.2	0.512 [0.40, 0.63]
instant_event_alert	4106	2.9	0.512 [0.44, 0.59]
snapshot_counting	3703	2.9	0.526 [0.43, 0.64]

[†] the only CI that excludes 0.5 — and it excludes it on the *wrong side*.

0.5 is chance, and eight of the nine CIs contain it. The obvious escapes were all checked and closed: not a clock offset (swept ± 10 s — no peak), not too tight a tolerance (± 10 s — unchanged), not a degenerate score (232–284 distinct values spanning 0.000–0.997).

Two details worth pausing on. **(i) Eight of nine AUCs are below 0.5.** Pure noise would scatter symmetrically about 0.5; a median of 0.455 is a systematic tilt, consistent with the predecessor’s finding that p_{hit} correlates with wall-clock time ($|r|$ up to 0.93) rather than with content. **(ii) dedup_counting’s CI excludes 0.5 on the wrong side** — its p_{hit} is significantly *anti*-correlated with events. That is worse than uninformative: thresholding it actively selects the wrong moments.

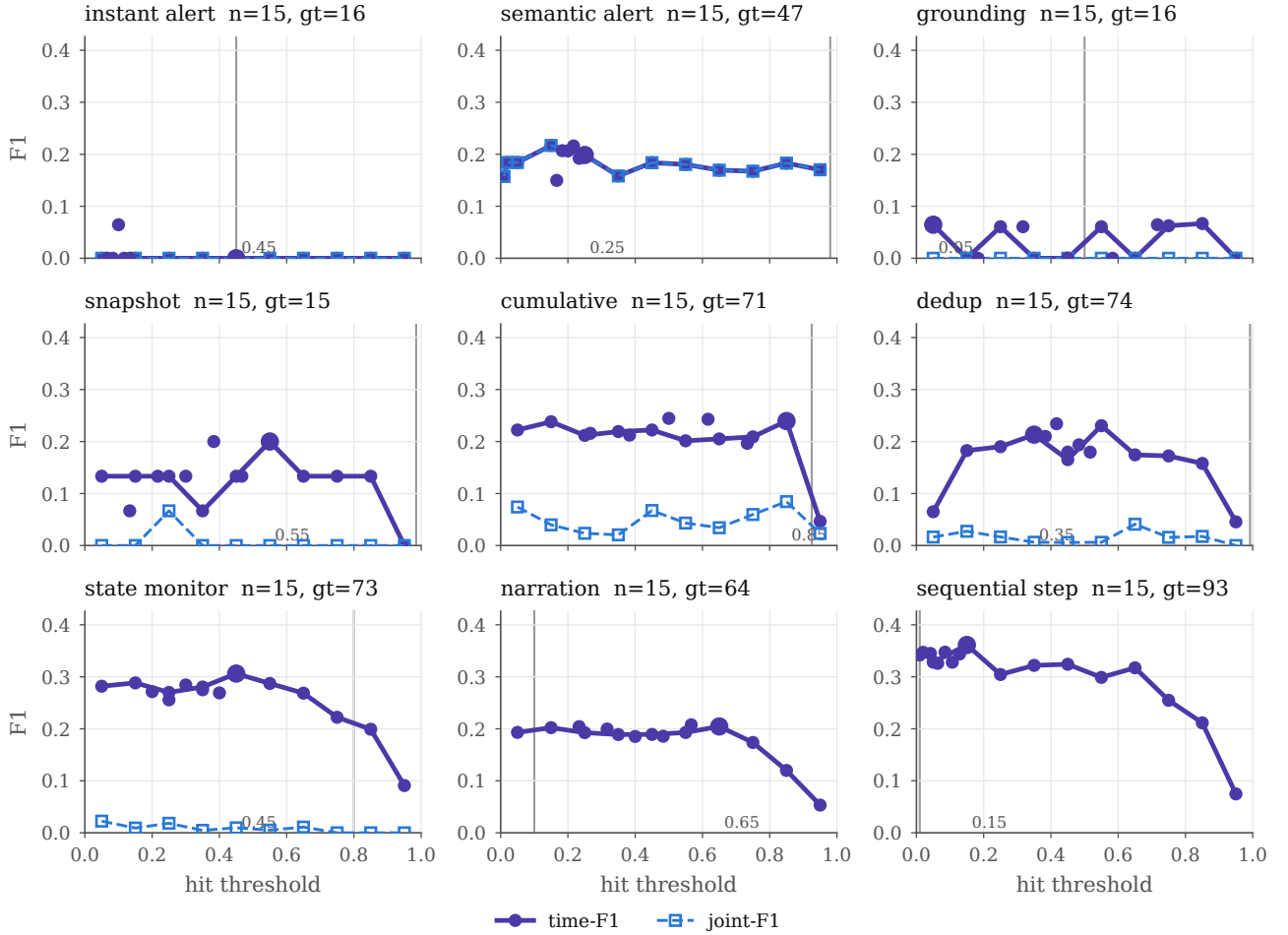


Figure 1: Pass 1, the wide grid. time-F1 against threshold for each task. The curves are flat across most of the range and collapse only at 0.95. A flat curve means the threshold is not selecting *moments* — it is selecting *volume*.

5.2 Finding 2 — no task’s fitted threshold survives its own bootstrap

Pass 1 completed (1,410/1,410, 94/94 cells) and the rule returned a winner for all nine tasks. The paired bootstrap then found:

- 9 of 9 tasks have a $\Delta F1$ -against-best-rival CI containing zero.
- Re-selection runs 25–61% against an 8–10% chance rate — `semantic_condition_alert` at just 6%, i.e. *below* chance.
- Two “fits” are artefacts: `instant_event_alert` scores 0.000 at every threshold, and `explicit_target_grounding`’s entire curve is one matched event or zero out of 16.

5.3 Finding 3 — the gate has one identifiable degree of freedom, not nine

All arms scored on the same videos at equal budget:

arm	time-P	time-R	time-F1	vs. best single global
<code>shipped_per_task</code>	0.151	0.356	0.2115	−0.0354, CI [−0.055, −0.016]
<code>global_0.5</code>	0.142	0.565	0.2271	−0.0198, CI [−0.039, −0.003]
<code>best_single_global</code> (0.15)	0.155	0.601	0.2469	—
<code>fitted_per_task</code>	0.163	0.582	0.2544	+0.0075, CI [−0.011, +0.027]

Nine fitted thresholds beat one global threshold by **+0.0075 with a CI straddling zero** — *in sample*, where the nine have nine free parameters to its one. Yet the single global value is *solid*:

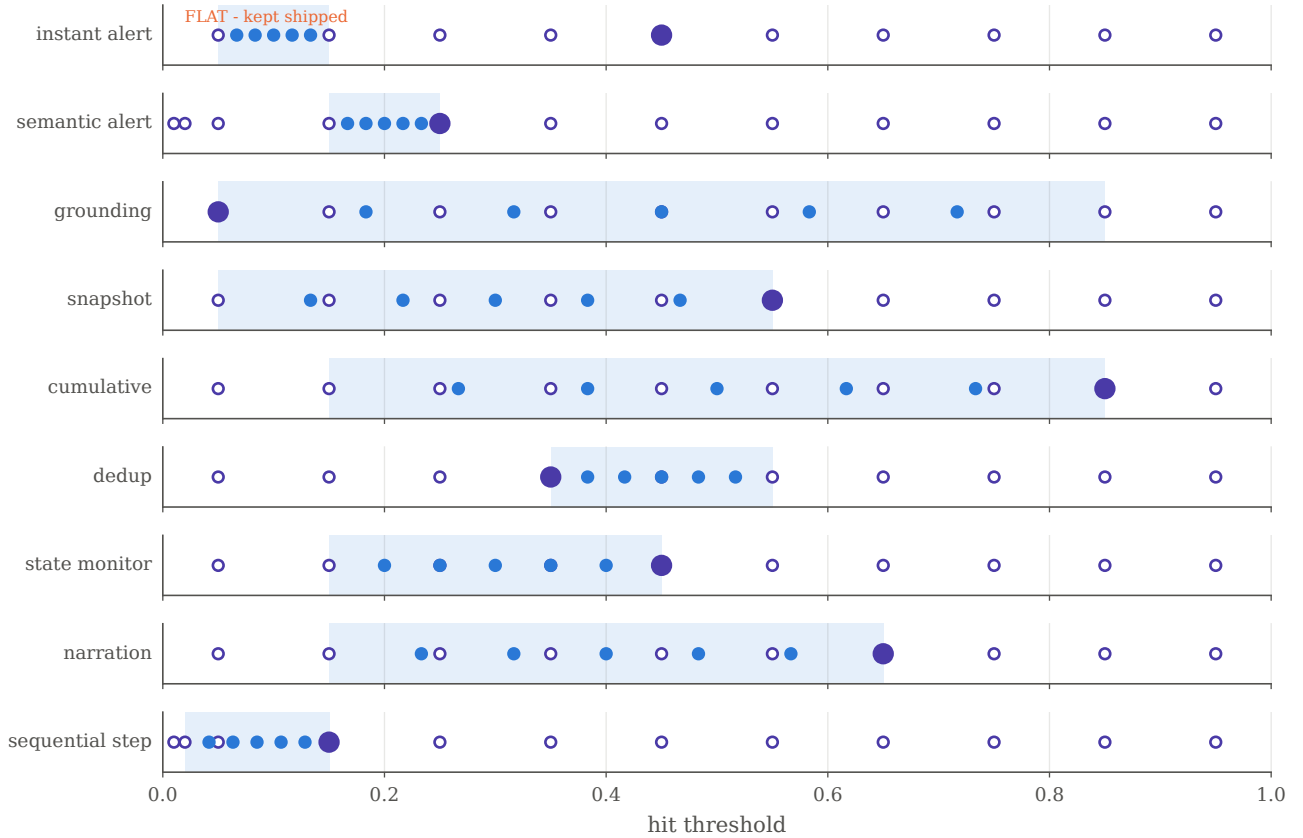


Figure 2: Pass 2, refined around each winner. The refined neighbourhoods span only 0.021–0.133 in F1, and on four tasks *every* cell falls inside the 0.03 noise band of the best. Refining did not resolve a winner; it confirmed there was no gradient to resolve.

re-running the sweep inside every bootstrap draw, **0.15 is re-selected in 94%** of them against 10% chance.

Pooled over 135 videos the operating point is identified; split nine ways over 15 videos each it is not. **That is a power problem, not a claim that the tasks are alike.** Note also that the *shipped* per-task thresholds are significantly *worse* than a flat 0.15 (-0.0354), and the damage is in recall (0.356 vs 0.601) — they were strangling the system. Replacing them is a real improvement even though the per-task *fit* is not.

5.4 Finding 4 — pass 2 confirmed a prediction registered before the data existed

Before pass 2 finished, a prediction was written down: the refined grid’s CIs would contain zero on *at least seven of nine* tasks. **The outcome was 9 of 9.**

The whole refined F1 span is 0.021–0.133, and on four tasks *every* cell sits within the noise band of the best. Two honesty notes are carried in the record: two rows report CI $[+0.000, +0.200]$ whose lower bound is *exactly* zero and therefore does not exclude it; and pass 2’s re-selection of 38–74% beats pass 1’s 25–61% only until you divide by chance (20% vs 8–10%) — **as a multiple of chance, pass 2 is no better.**

5.5 Finding 5 — the fit does not overfit

Stage 3’s 2,700 samples *contain* the 135 fitting videos, making the headline $\approx 5\%$ in-sample. Rescoring with those ids removed — free, the same predictions filtered — gives an in-sample bias of **+0.0007 gross**, two orders of magnitude inside the noise band. Whatever else is true of this fit, it generalises. The negative result is about *identifiability*, not overfitting.

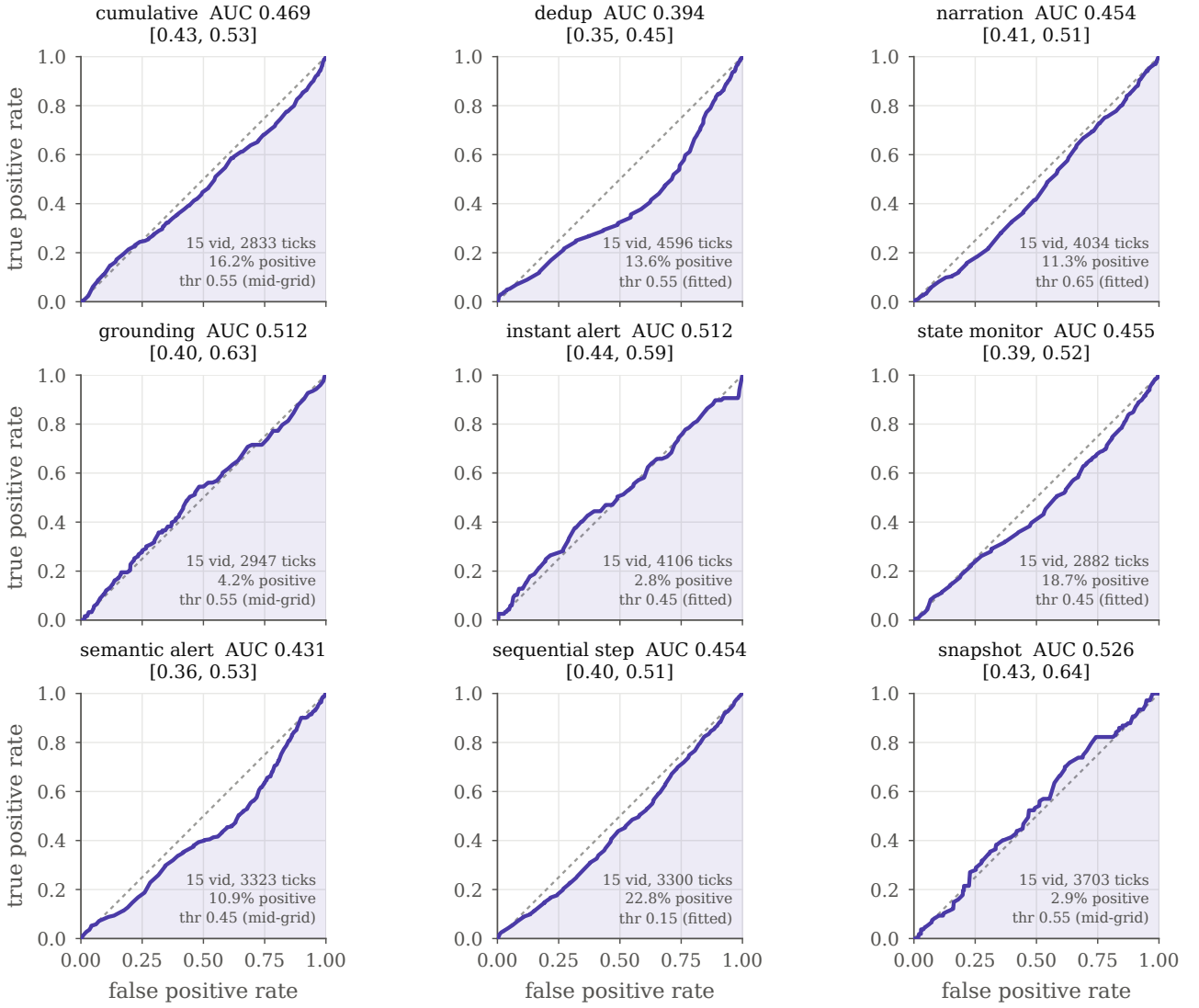


Figure 3: Finding 1, drawn. ROC of per-tick p_{hit} against $\pm 3s$ event-adjacency, one panel per task, with bootstrap CI, tick count, positive rate, and the threshold used. Every curve hugs the chance diagonal; `dedup_counting` bows *below* it. Panels state their own n so a 15-video panel cannot be mistaken for a 300-video one.

6 Stage 3: the full benchmark, per task

At 2,563 of 2,700 banked (95%), judge offline:

task	n	n_{gt}	n_{emit}	emit/gt	t-P / t-R	time-F1
sequential_step_instruction	298	1685	5139	3.05	0.251 / 0.765	0.3778
realtime_state_monitor	271	1200	6197	5.16	0.143 / 0.739	0.2398
cumulative_counting	288	1379	3291	2.39	0.165 / 0.393	0.2321
event_narration	283	1236	5557	4.50	0.138 / 0.619	0.2252
dedup_counting	285	1156	4713	4.08	0.127 / 0.519	0.2045
semantic_condition_alert	292	956	4322	4.52	0.123 / 0.557	0.2016
explicit_target_grounding	287	316	307	0.97	0.104 / 0.101	0.1027
snapshot_counting	279	279	279	1.00	0.065 / 0.065	0.0645
instant_event_alert	280	366	280	0.77	0.068 / 0.052	0.0588
gross (micro)	2563	8573	30085	3.51	0.156 / 0.546	0.2423
macro over tasks						0.1897

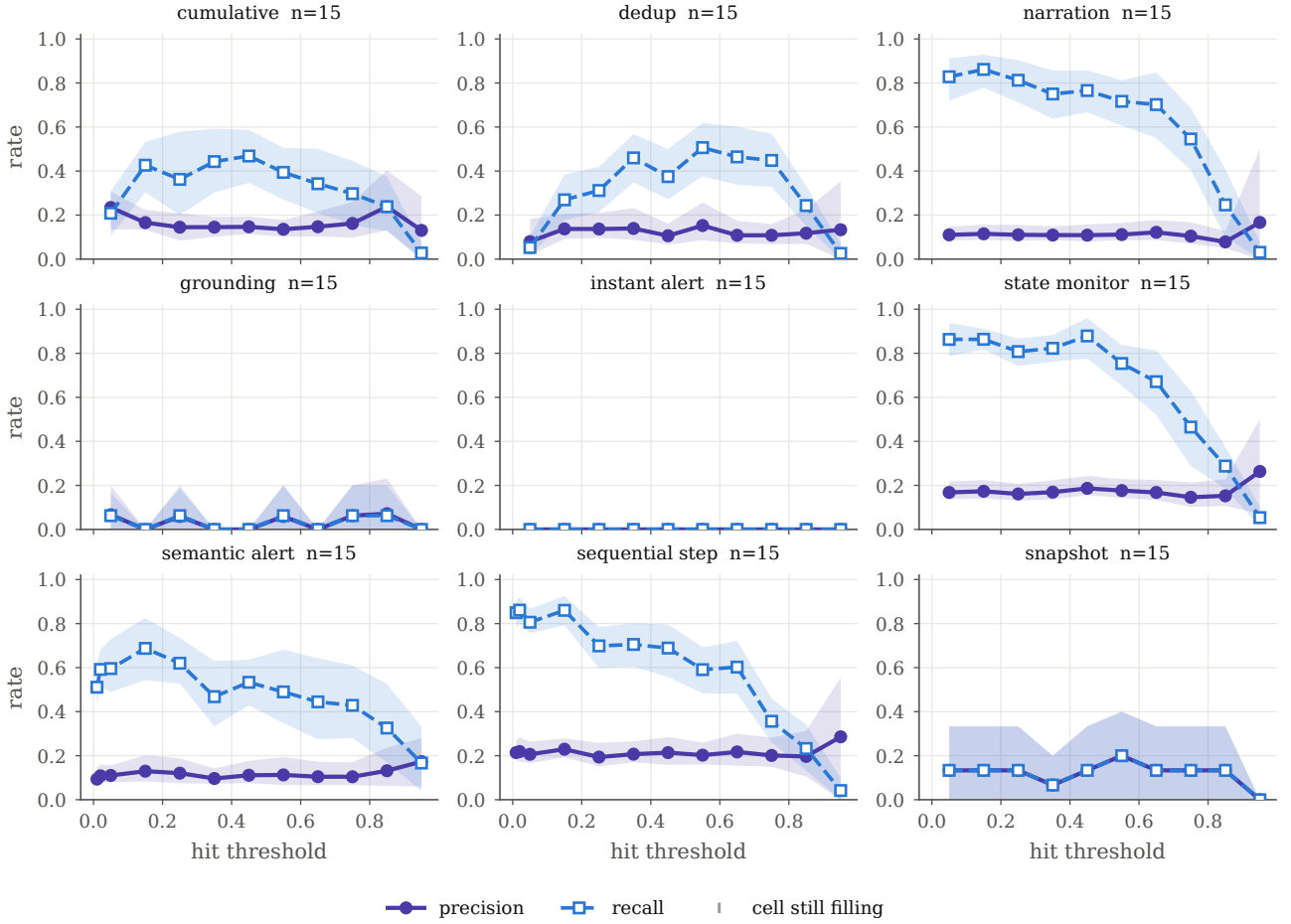


Figure 4: Precision is flat while recall falls. Sweeping the threshold moves recall and leaves precision alone — exactly what AUC ≈ 0.5 predicts, and arrived at through a completely separate code path from the ROC above.

6.1 Two failure regimes, not one

Regime A — over-emission (six tasks). 2.4–5.2 emissions per event. Precision, not recall, is what limits these. Gross precision is 0.156 against recall 0.546: the system finds the moments and buries them in false positives.

Regime B — one emission per video (three tasks). These carry 300 s and 600 s refractories inherited from the vision-only fit. `snapshot_counting` is the clean case: exactly 1.00 ground-truth event per video, exactly 1.00 emission, and F1 **0.0645** — so the single emission lands within ± 3 s only about 6% of the time. The *count* is perfect; the *moment* is wrong.

The gate is nearly exhausted, and the intuitive fix is backwards. With precision flat across the grid, $F_1 = 2pr/(p+r)$ rises monotonically as $r \rightarrow 1$ to a ceiling of $2p/(p+1) = \mathbf{0.272}$. We sit at 0.242. So every remaining threshold adjustment in existence is worth ≤ 0.03 — inside the noise band — and *cutting* the over-emission would **lower** F1, because precision will not rise to compensate.

6.2 Against the vision-only predecessor

	emits per event	micro time-F1	macro time-F1
system_3 (Qwen3-VL-8B, vision), $n=931$	1.48	0.2684	0.2565
omni (Qwen2.5-Omni-7B, A+V), $n=2563$	3.51	0.2423	0.1897

We win on 2 of 9 tasks. Micro-averaged we trail by 0.026 (borderline against the noise band);

Fig 7 Every interval crosses zero, in both passes

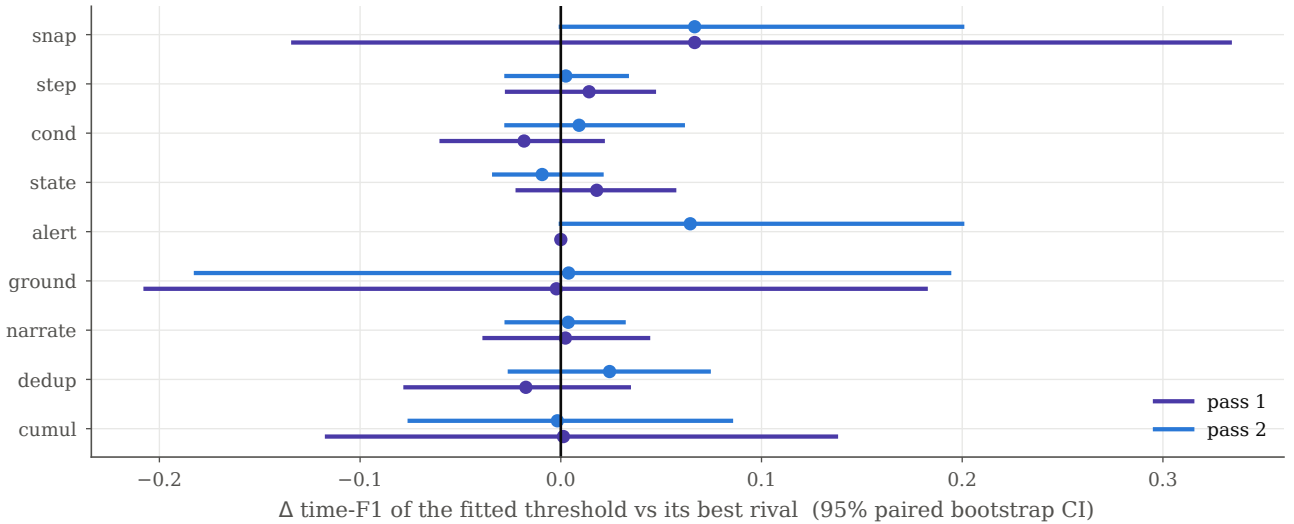


Figure 5: Findings 2 and 4 in one picture. Each task’s Δ time-F1 of the fitted threshold against its best rival, with 95% paired-bootstrap CI, for both passes. **Every interval crosses zero.** Pass 2’s intervals (blue) are narrower than pass 1’s, but narrower is not the same as excluding zero. The two pass-2 intervals whose left end sits exactly on zero are the degenerate tasks matching 0–2 events out of 15–16; an inclusive bound of exactly zero does not exclude zero.

macro-averaged by 0.067, which is not. The micro figure flatters us because it is dominated by the tasks we do adequately.

The two systems fail in opposite directions. The predecessor emits $1.48\times$ per event at precision 0.225 / recall 0.333; we emit $3.51\times$ at 0.156 / 0.546. It is conservative and accurate; we are talkative and vague. Our two wins are both **level-mode** short-refractory tasks — exactly where talking more pays.

And where we lose most, it simply gets more attempts. On the three regime-B tasks the predecessor emits 2.85 / 3.07 / 1.33 times per event where our inherited refractories permit ≈ 1.00 . Inverting $F_1 = 2tp/(k+1)$ for one event and k emissions: it converts $\approx 53\%$ of **snapshot_counting** videos into a hit with 2.85 attempts ($\approx 24\%$ per attempt); we convert 6% with one. So *both* factors are real — our per-emission timing is $\approx 4\times$ worse *and* we take a third as many attempts.

7 Limitations — read before quoting any number

1. **Only one of three coupled knobs was refitted.** `mode` and `refractory_s` are inherited unchanged. The evidence above says they are coupled to the threshold, and §6.1 shows the cost concretely: the 300s/600s refractories permit ≈ 1 emission per video on the three worst tasks.
2. **There is no out-of-sample nine-versus-one comparison.** A control arm (flat global 0.15 over all 2,700) was built, preflighted, then **cancelled before launch**. The only fitted-vs-global number is the in-sample +0.0075, and it must always be cited as *in sample, $n=135$, nine free parameters against one*. Two predictions registered for that arm are **withdrawn, not resolved**.
3. **15 videos per cell is underpowered**, and that is the direct cause of Findings 2–4 rather than an incidental weakness.
4. **content_acc and joint_f1 are withheld** throughout. No LLM judge has produced a verdict in any run of this fork: the workers `unset` the API keys deliberately, and the configured model (`gemini-3-flash`) 404s when tried. Seven of nine tasks are content-scored by deterministic parsers (`count`, `position`, `state`, or `time_only`); the two free-text tasks are unjudged, so the *pooled* figure cannot be formed. The true joint-F1 lies between the lower bound 0.038 and the timing ceiling 0.242. No comparison with published baselines (MiniCPM-o 4.5 at 0.209 *joint-F1*) is available — and they used a different judge on a different subset.

Fig 6 Stage 3, per task: score and the emission ratio behind it

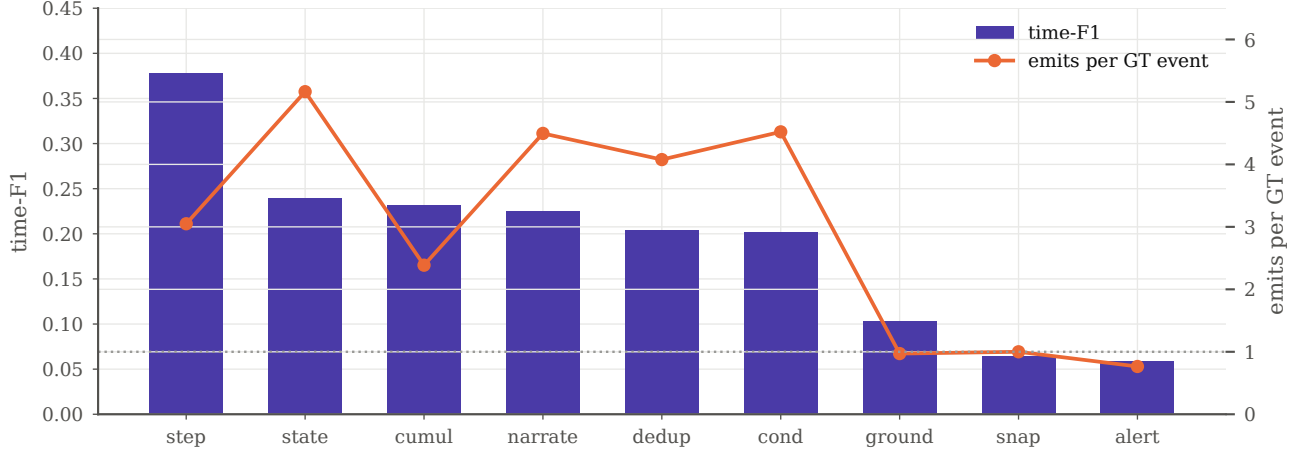


Figure 6: The result, and the mechanism beside it. Bars are per-task time-F1; the line is emissions per ground-truth event, with the dotted reference at 1.0. The two regimes are visible at a glance: six tasks emit 2.4–5.2× too often, three emit about once per video.

5. **The eval is not bit-reproducible.** Treat <0.03 as noise, always.

6. **The deepest one.** A separate diagnosis found the controller’s ICL schema *collapsing* on this backbone: a mean of **21.7 generated tokens against a 300-token budget**, with the self-pacing fields appearing in 20 ticks out of 78,828 and **answer** present on only 51%. The boolean this entire study thresholds is produced by that generation step. So Finding 1’s $AUC \approx 0.5$ is plausibly **an artefact of a broken upstream stage rather than a property of the task**. The findings remain correct *about the system as it ran*; what they predict about a repaired system is an open question.

8 Conclusions, and what to do next

Per-task threshold fitting did not work, and this study establishes *why*: not because the tasks are alike, but because the signal being thresholded carries no timing information, and because 15 videos per task cannot identify nine parameters. A single global threshold (0.15) is the only operating point the data identifies, and it is identified well (94% re-selection against 10% chance).

Worth carrying forward:

- A negative result established by **four independent code paths** — ROC, flat precision curve, paired bootstrap, refined grid — with predictions registered before the data and *withdrawals recorded as visibly as confirmations*.
- The fit **generalises** (+0.0007 in-sample bias).
- The shipped thresholds are **measurably harmful** versus a flat 0.15.

The next move is not a better gate — with precision flat there is ≤ 0.03 left in it. Ranked by evidence:

1. **Use `event_time_s`, which is free.** The model already names an onset time on 41% of ticks and the gate discards it, timestamping emissions at the crossing tick instead. The data is already banked; testing costs no GPU.
2. **Make the controller finish its schema** (constrained decoding, or a smaller schema this backbone can complete).
3. **Refit mode and refractory**, which the head-to-head shows is worth more than the ceiling analysis alone suggested — it binds on the three worst tasks.

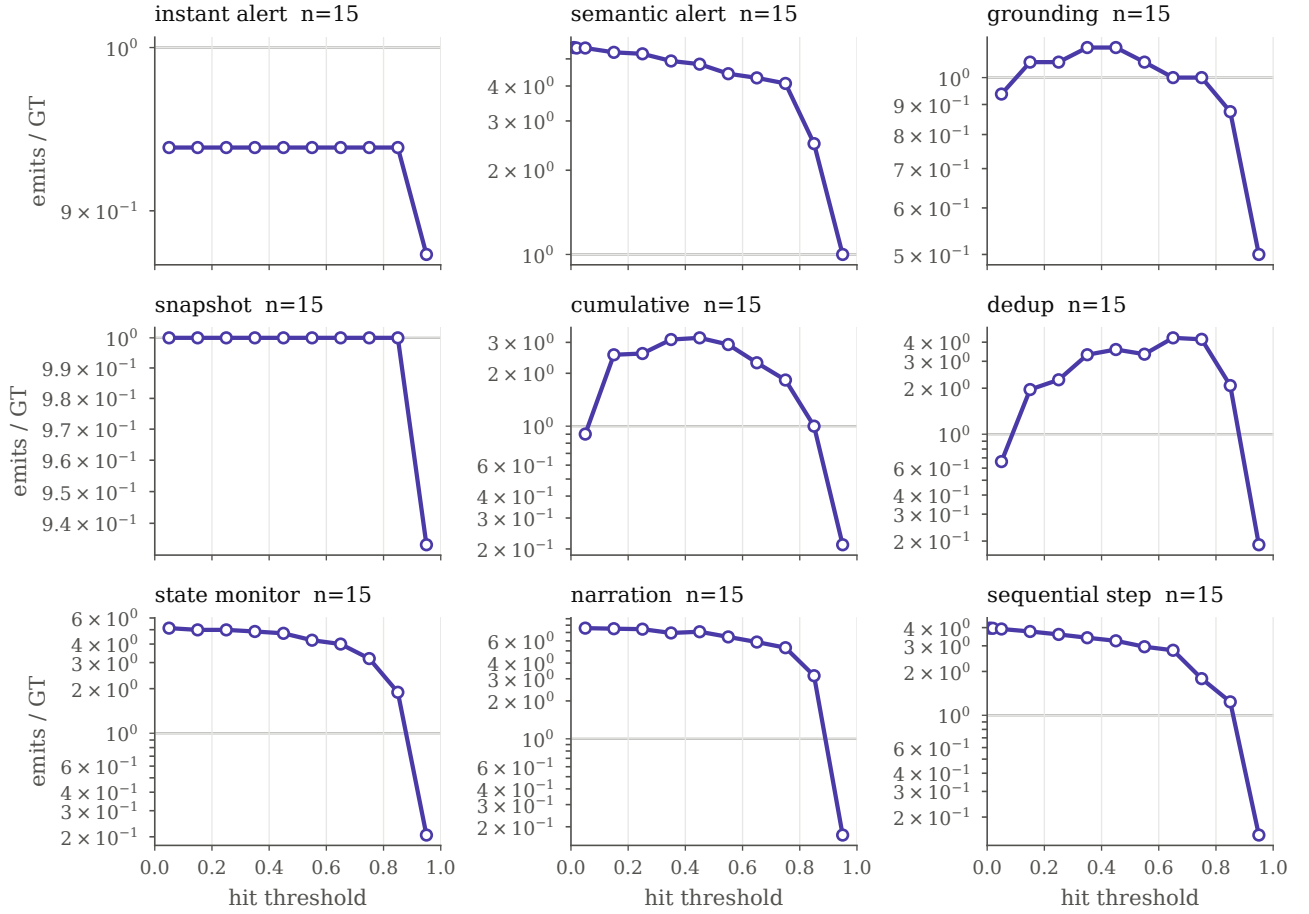


Figure 7: Emission calibration. How many times the system speaks against how many events exist. A well-calibrated gate would track the diagonal.

4. Restore a judge. Every content number is blocked behind it.

Evidence. Every figure and number is recomputed from banked predictions and re-derivable without a GPU: AUC_TABLE.json and PERCEPTION_AUDIT.json (Finding 1), FIT_NOISE_AUDIT.json / _P2.json (2, 4), ABLATION.json (3), STAGE3_PARTIAL.json (§6), FINAL_THRESHOLDS.json, STAGE3_CONFIG_APPLIED.diff (what stage 3 actually loaded). Figures are built by lib/figs.py, lib/figs_auc.py, lib/figs_stage3.py. The full running record, including every deviation from the original spec and why, is THRESHOLD_FIT_RUNBOOK.md §2.1–2.10.

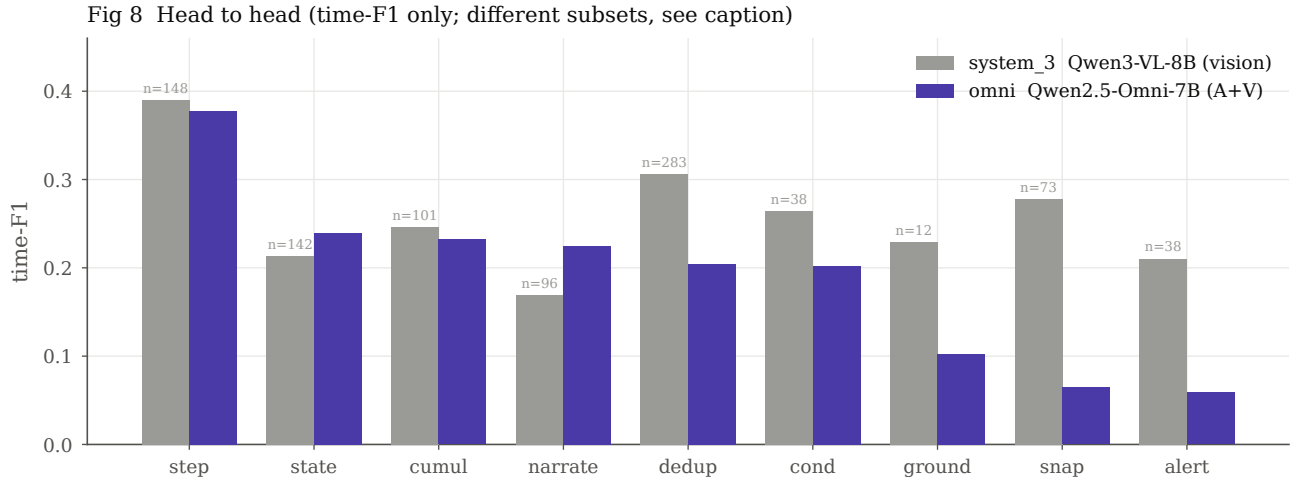


Figure 8: Head to head, time-F1 only. n is printed over each predecessor bar because it is uneven and small *exactly where that system wins most* (ground $n=12$, alert $n=38$) — those two comparisons carry the least weight in the chart, not the most. Different subsets, backbones and modalities: a signpost, not a controlled comparison.